

Dhruvil Satasiya

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Surat, Gujarat-394101

OBJECTIVE

Machine Learning Engineer specializing in mechanistic interpretability and adversarial robustness. Proven track record of translating theoretical alignment research into optimized PyTorch infrastructure. Graduated (B.Tech) and available for immediate, full-time remote deployment.

EDUCATION

• P P Savani University, Surat

2022–May 2026 (Graduated)

B.Tech in Information Technology

Relevant Coursework: Artificial Intelligence, Optimization Techniques, Big Data Analytics

• Indian Institute of Technology Madras (IITM), Chennai

2023– Apr 2027

BS in Data Science

Relevant Coursework: Machine Learning Practice, Statistics, Data Structures, DBMS, Python Programming

Note: This is a flexible, asynchronous distance-learning degree that requires zero daytime hours and does not impact full-time employment availability.

PUBLICATIONS

Dhruvil S., F. Sojitra, R. Chauhan. “Through the Bottleneck: How Multi-head Latent Attention Separates Content from Position in Language Models.” *arXiv preprint arXiv:2607.23054*, 2026. [arXiv Link] | [GitHub Repo]

INTERNSHIP EXPERIENCE

• AI Research Intern

June 2026-Present

Remote

AI Institute of South Carolina

- SLM Pretraining Pipeline: Architecting the end-to-end pretraining and alignment infrastructure for a novel Small Language Model (SLM).
- Data Infrastructure: Engineered the initial large-scale dataset tokenization and processing pipeline to prepare the raw corpus for the distributed pretraining run.

• Independent ML Researcher

Dec 2025-Present

- Mechanistic Interpretability: Conducted the first comprehensive mechanistic interpretability study on Multi-head Latent Attention (MLA), utilizing SVD and linear probing on a 114M-parameter model to demonstrate how the shared c_{KV} bottleneck preserves content, discards position, and fundamentally reshapes circuit topology.
- Adversarial Robustness: Developed open-source, from-scratch PyTorch implementations of advanced white-box (Greedy Coordinate Gradient) and black-box (Prompt Automatic Iterative Refinement) red-teaming algorithms.

• AI/ML Intern

June 2025-Nov 2025

On-site

10Turtle

- Designed and deployed production LLM systems with improved latency, cost efficiency, and response quality.
- Ran controlled experiments across model configurations to optimize performance in real-world settings.
- Built containerized inference services with automated logging, health checks, and monitoring to ensure >99% uptime.
- Trained and deployed YOLO-based object detection models for multi-class image analytics, optimizing data augmentation and inference pipelines.
- Converted experimental LLM workflows into production-ready systems using reproducible evaluation and logging pipelines.

• SDE Intern

Jan 2024-May 2025

Remote

Ashirwad Infotech

- Deployed a fine-tuned LLM on GPU instances using vLLM, exposing inference APIs for low-latency serving.
- Optimized inference by resolving GPU OOM issues and evaluating 4-bit/8-bit quantization and dynamic batching configurations to improve throughput and memory efficiency.
- Fine-tuned an instruction-following LLM using SFT with LoRA/QLoRA for parameter-efficient adaptation.
- Prepared chat-style instruction datasets, including formatting, tokenization, and prompt template construction using Hugging Face and PyTorch.
- Managed the complete fine-tuning workflow from data preparation to evaluation and adapter merging.

ADDITIONAL PROJECTS

• JEPA for Text (LeCun-inspired)

Source Code

- Implemented predictive representation learning system to study latent structure in language.
- Investigated embedding geometry and model stability without reconstruction objectives. **Tech:** Python, PyTorch, Transformers, Scikit-learn, Matplotlib.

TECHNICAL SKILLS

- **Programming:** Python
- **Libraries/Frameworks:** PyTorch, Transformers, Accelerate, PEFT, NumPy, Pandas, Matplotlib, Seaborn
- **NLP & LLM Tools:** vLLM, Weights & Biases, LlamaIndex, LangChain, OpenAI APIs
- **ML Techniques:** Model Fine-tuning, A-B Testing, Object Detection, Sequence Modeling, Experiment Design
- **Deployment:** Docker, Logging/Monitoring, API Integration, Git
- **Soft Skills:** Clear technical communication, research mindset, rapid prototyping, teamwork

CERTIFICATIONS

- **Reinforcement Learning** NPTEL (IIT Madras) | Elite Certification *Jan-Apr 2025*
- **AWS Academy Cloud Foundations** AWS Academy *Mar 2025*